



Getting Started with Managed Services

Question: Do you want to run apps in the cloud like you do in a data center?

- **Alternative:** Or are there better cloud-native approaches?

You should understand some terminology:

- **IaaS** (Infrastructure as a Service)
- **PaaS** (Platform as a Service)
- **SaaS** (Software as a Service)

Ready for a journey to understand these?

Remember these $IAAS < PAAS < SAAS$, are service models, for giving some services to us

As here we developers handles everything by own

Less control & responsibility

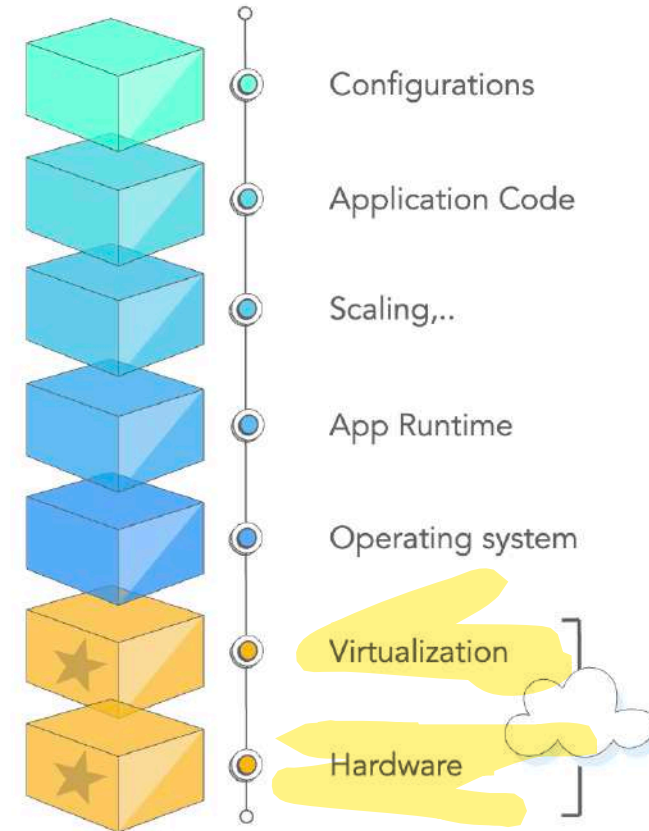


More control & responsibility



Example 1A: Setting up an App - IaaS Approach

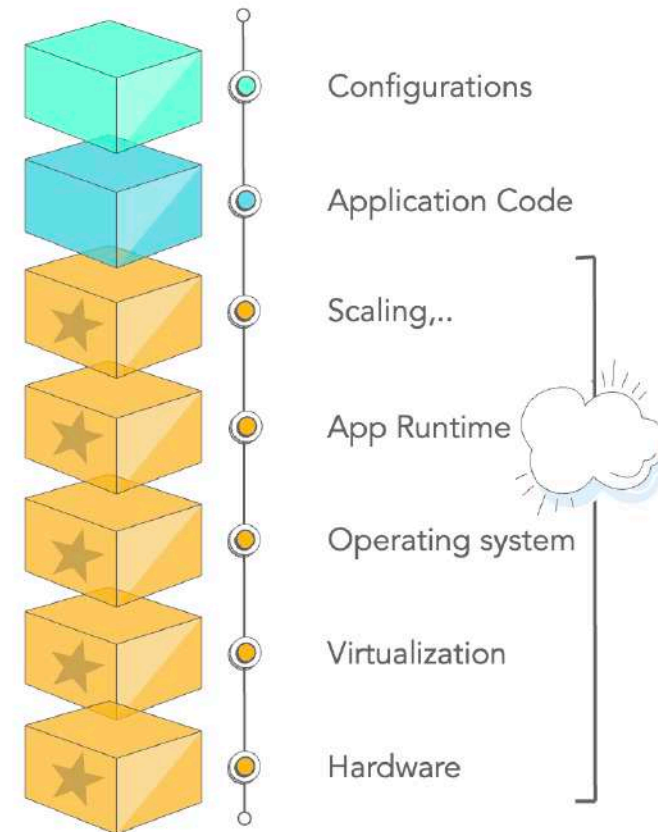
- 1: Provision VMs – Create VMs on cloud
- 2: Install Software – Install OS, runtimes etc
- 3: Deploy Code – Upload code and configure your application manually
- 4: Configure Ops – Set up load balancing and auto scaling yourself
- 5: Maintain – You are responsible for OS patching, runtime upgrades and server health





Example 1B: Setting up an App - PaaS Approach

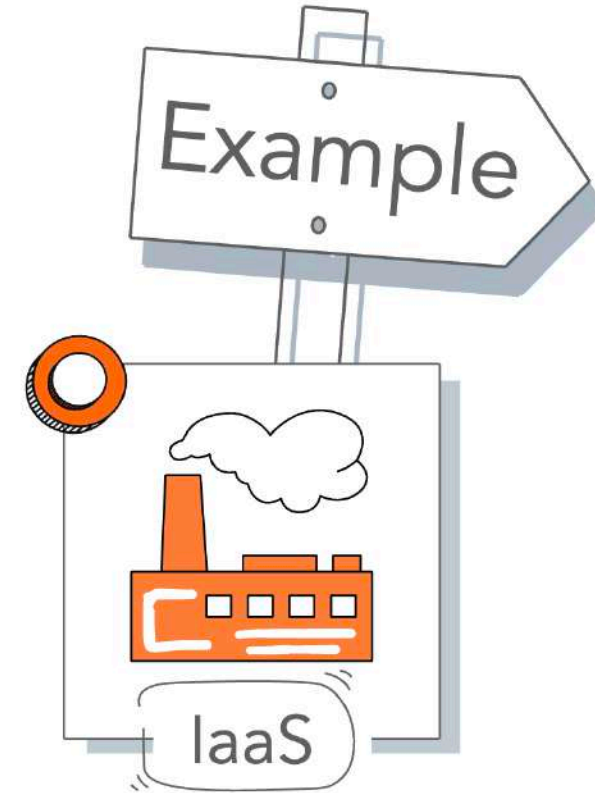
- 1: Pick a Platform** – AWS Elastic Beanstalk, Azure App Service, or Google App Engine
- 2: Upload Code** – Upload your code artifact
- 3: Automatic Setup** – Platform provisions VMs, installs OS, and starts app
- 4: Built-in Ops** – Scaling and Load Balancing are easily configurable
- 5: Zero Maintenance** – No Manual OS patching or server management required





Example 2A: Setting up a SQL Database - IaaS Approach

- 1: Provision VMs – Create VMs to host your database
- 2: Install Database – Set up MySQL, PostgreSQL, or any DB you need
- 3: Configure Storage – Attach disks
- 4: Set Up Ops – Manually configure backup jobs and replication
- 5: Maintenance – Manage scaling, patches, and monitor performance

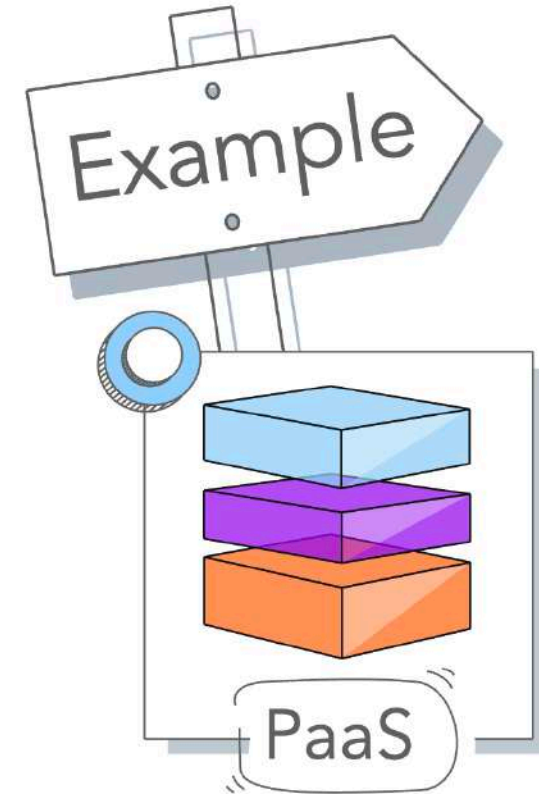




Example 2B: Setting up a SQL Database - PaaS Approach

- 1: **Pick a Service** – AWS RDS, Azure SQL Database, or Google Cloud SQL
- 2: **Choose Type** – MySQL, PostgreSQL, SQL Server, etc
- 3: **Configure** – Decide CPU, memory, and storage size
- 4: **Enable Ops** – Turn on built-in backups & high availability
 - **Ready** – Database is ready to use

Focus: Data and queries, **NOT** servers or maintenance





What is IaaS?

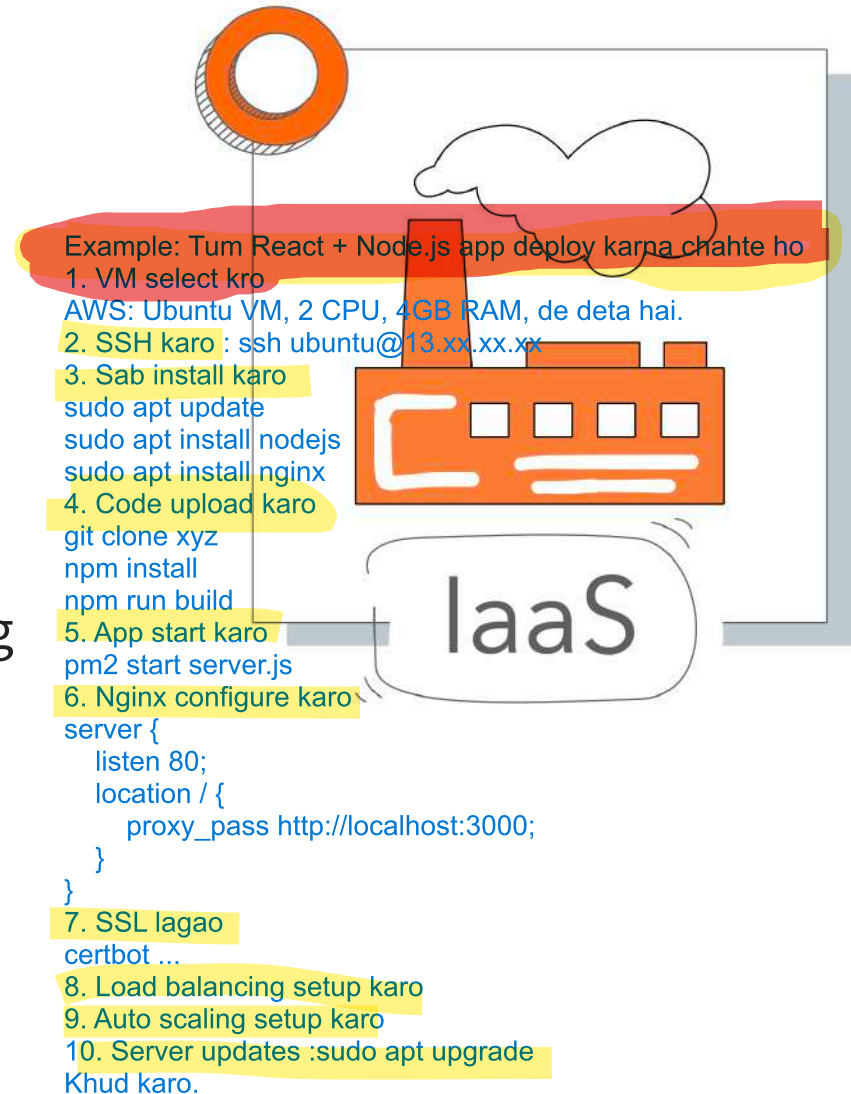
IaaS: Infrastructure as a Service

- **Concept:** Use only infrastructure from the cloud provider

Customer Responsibility: High

- **Components:** OS, runtime, and system software
- **Operations:** Availability, durability, and scaling

Summary: You manage almost everything except the hardware





What is PaaS?

PaaS (Platform as a Service): Use a complete platform provided by the cloud

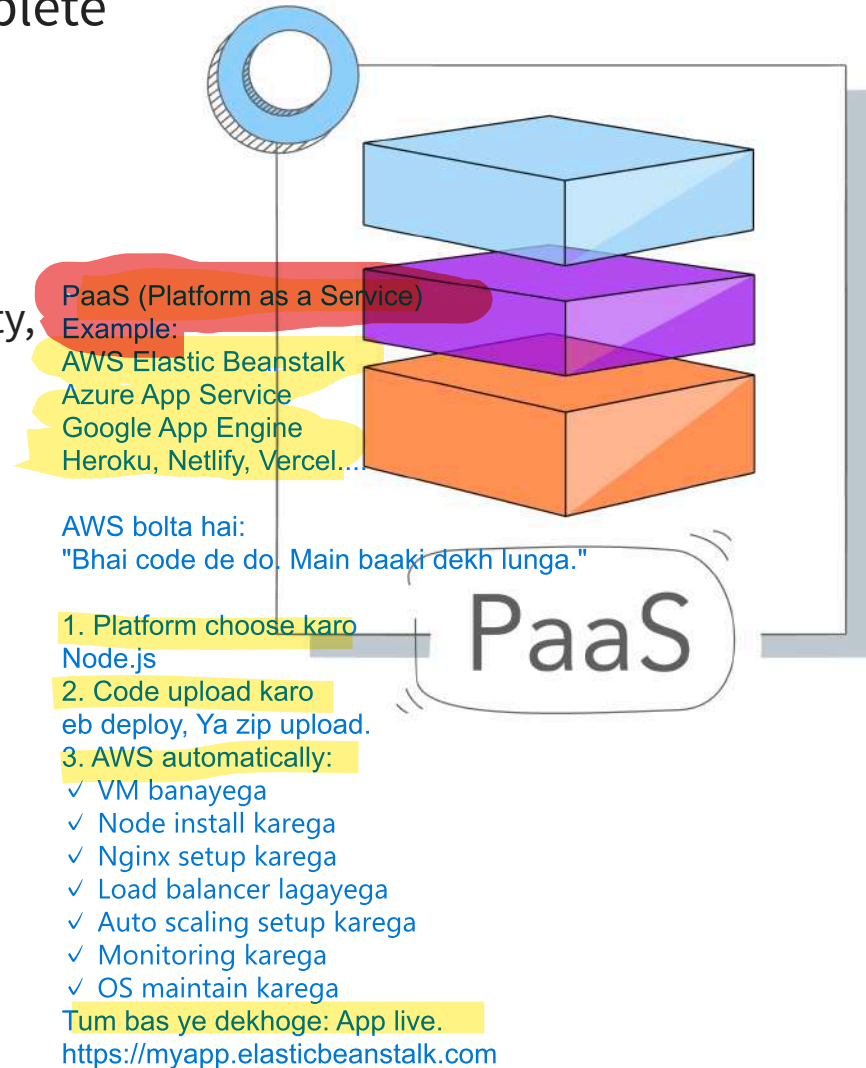
Cloud Provider Responsibility: High

- **Components:** OS, runtime, availability, durability, scaling

Customer Responsibility: Low

- **Focus:** Code, data, and configuration

Summary: Cloud provider manages the routine work





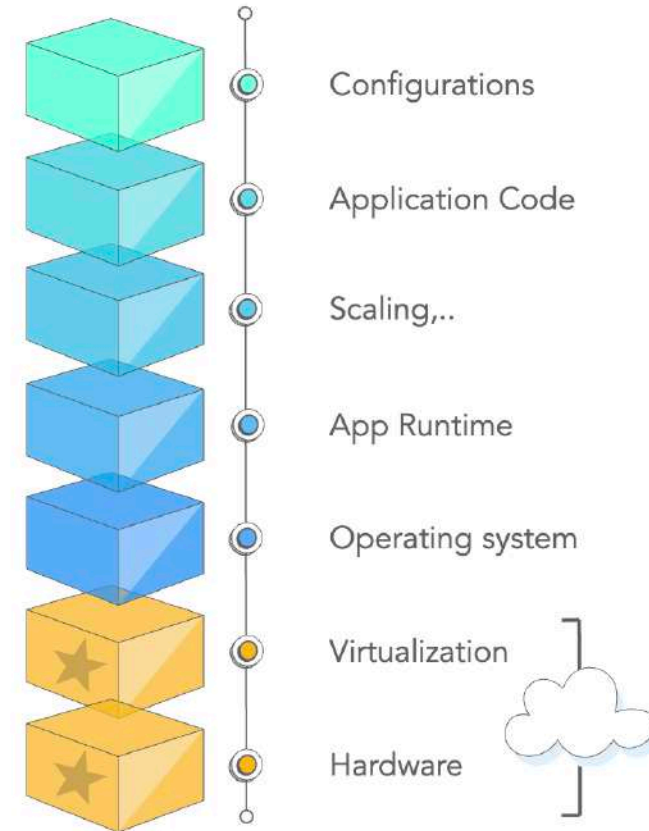
Who is Responsible for What? - IaaS

Cloud platform: The Basics

- Virtualization, Hardware and Networking

Customers of cloud: Everything Else

- **Application:** Code and Runtime
- **Operations:** Load balancing, Auto scaling, Availability
- **Database:** Software, upgrades, configuration (tables, views,..), and data
- **OS:** Upgrades and patches





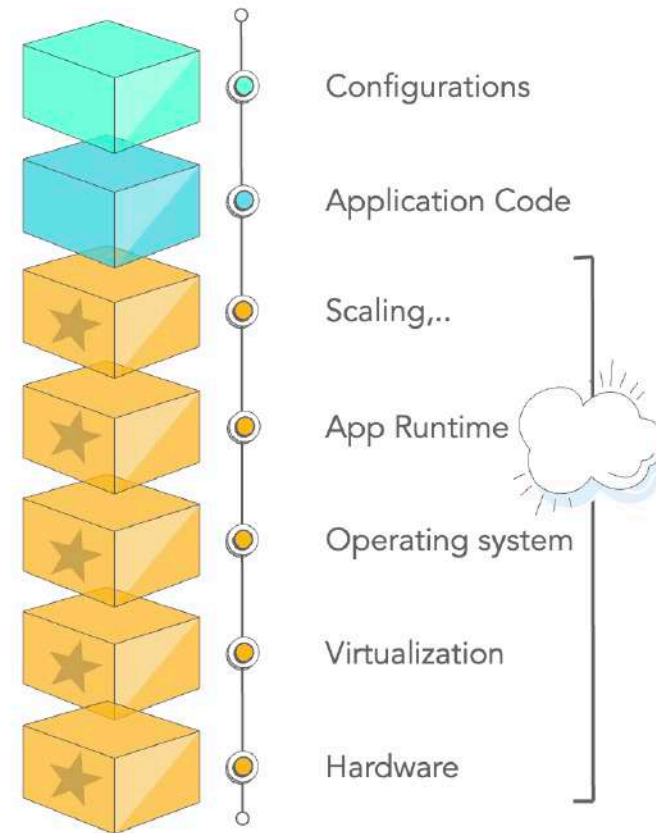
Who is Responsible for What? - PaaS

Cloud Platform: Heavy Lifting

- **Infrastructure:** Virtualization, Hardware, Networking, *OS: Upgrades and patches
- **Platform:** Runtime, Database software, Scaling, Availability, Durability, Load Balancing

Customer: Business Logic

- **Application:** Code & configuration
- **Database:** Schema, Tables, and Data
- **Restricted Access:** Usually no access to OS or VM instances





PaaS Services Can Do a Lot More

Scope: PaaS is NOT limited to running apps and databases

Managed Services: Includes common building blocks

- **Storage:** Amazon S3, Azure Blob Storage, Google Cloud Storage
- **Analytics:** BigQuery, Azure Synapse, AWS Redshift

Goal: Let developers focus on logic and innovation

- **Benefit:** Skip the setup and maintenance



Personal Notes on managed services at last



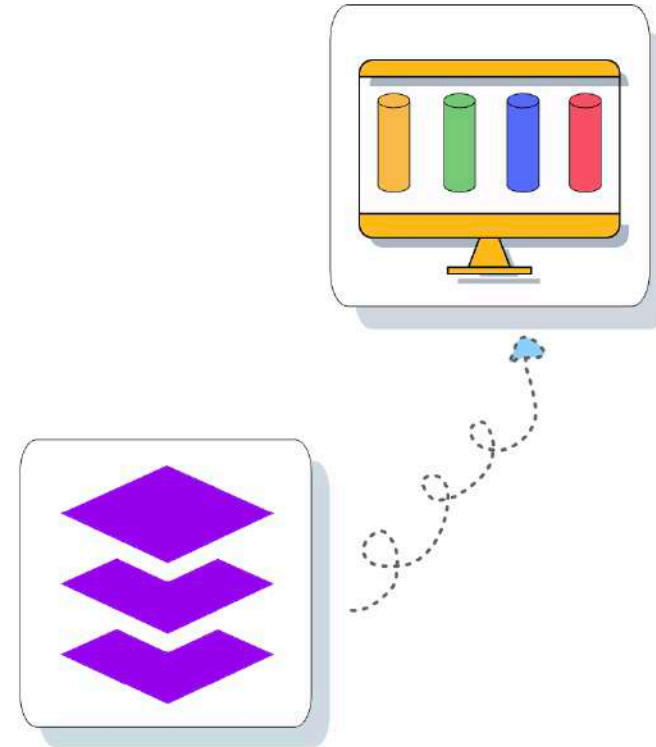
From Platforms to Complete Applications

IaaS: You manage most things yourself (infra, scaling, app, data)

PaaS: The platform helps you focus on your application and data

SaaS: You use complete software - no setup needed!

Next Step: Let's explore SaaS (Software as a Service)





What is SaaS (Software as a Service)?

Definition: Centrally hosted software

- **Hosting:** Mostly on the cloud

Billing: Offered on a subscription basis (Pay as you go)

Examples:

- **Email:** Gmail, Outlook 365
- **Office tools:** Microsoft Office 365, Google Docs
- **Calendar:** Google Calendar





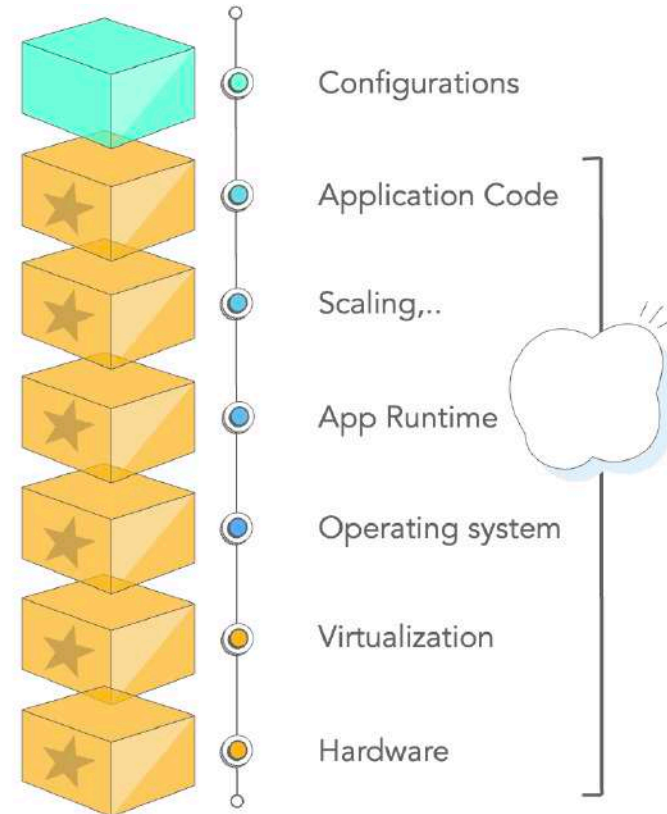
Who is Responsible for What? SAAS

Service Provider: Almost Everything

- **Infrastructure:** OS, Runtime, Networking
- **Application:** Code updates and maintenance
- **Operations:** Auto scaling, Availability, Load balancing

Customer: Usage

- **Content:** Documents, emails, sheets
- **Configuration:** Users, permissions etc





Quick Recap - IaaS vs PaaS vs SaaS

IaaS: Platform provides infra

- **You:** Manage all other layers

PaaS: Platform handles infra, setup, and scaling

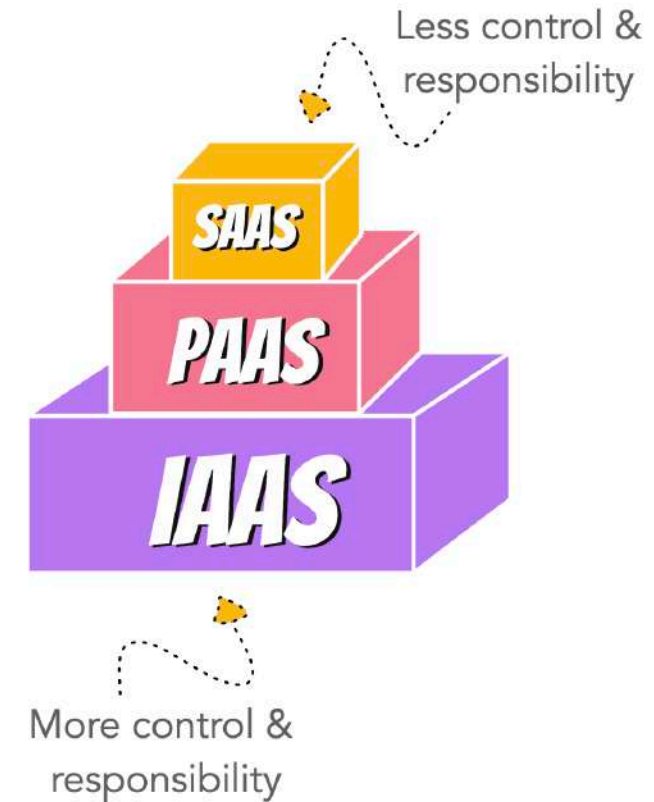
- **You:** Handle code, configuration, and data

SaaS: Platform provides everything

- **You:** Just use the app

Trend: Moving from IaaS → PaaS → SaaS

- **Control:** Decreases
- **Ease of Use:** Increases





Service Categories - Scenarios

Scenario	Solution
IaaS or PaaS or SaaS: Deploy Custom Application in Virtual Machines	IaaS
IaaS or PaaS or SaaS: Deploy a Database in Virtual Machines	IaaS
IaaS or PaaS or SaaS: Using Gmail	SaaS
IaaS or PaaS or SaaS: Using a Managed Service to setup a database	PaaS
True or False: Customer is responsible for OS updates when using PaaS	False
True or False: Customer is completely responsible for Availability when using PaaS	False
True or False: In PaaS, customer has access to VM instances	False
True or False: In PaaS, customer can customize OS and install custom software	False
True or False: PaaS services only offer Compute services	False
True or False: In PaaS, customer can configure hardware needs (memory, cpu etc)	True



Choosing Between IaaS, PaaS, and SaaS - Scenarios

Scenario	Solution
Legacy Migration: You must migrate a 10-year-old app that requires a specific, older version of the OS with customization	IaaS
Rapid Development: A startup wants to deploy a new NodeJS web app immediately. They have no Ops team and don't want to manage servers	PaaS
Productivity: You need email, calendar, and document collaboration tools for 500 employees starting tomorrow	SaaS
Custom Database: You need to run a specialized database engine that needs customization of Operating System	IaaS
CRM Needs: Your sales team needs a tool to track leads and customers. You want zero maintenance overhead	SaaS

Hrr Cloud provider ke paas, hrr cheej ke liye managed services hai.

Managed Service ka actual matlab kya hai?

Managed service = "Ye kaam tum mat karo, hum kar denge."

Example 1: Database

Without Managed Service

Tum EC2 (VM hai) lete ho. Fir:

Install Ubuntu -> Install PostgreSQL -> Configure Users -> Setup Backups -> Setup Replication -> Monitor CPU -> Upgrade PostgreSQL ->Handle Failures

Sab tum karoge.

Managed Service (RDS)

AWS bolta hai: Kaunsa DB chahiye?
PostgreSQL?, MySQL, MariaDB, SQL Server, Tum select karte ho.

Then AWS: DB install Karega, Backups lega, Monitoring Karega, Patching Karega, Replication setup Karega, Failover karega

Tum bas use karoge: db.connect(...)

Example 2: Storage

Without Managed Service

Tum EC2 (VM) lete ho. Fir:

Attach Disk → Manage Files → Handle Replication → Ensure Durability → Monitor Storage Usage → Expand Capacity → Handle Failures

Sab tum karoge.

Managed Service (S3)

AWS bolta hai: "File upload karo." Bas.

Then AWS: Storage manage karega → Replication karega → Durability ensure karega → Availability maintain karega → Scaling automatically karega → Hardware failures handle karega

Tum bas use karoge: s3.upload(...)

Example 3: Analytics

Maan lo tumhe 100 TB data analyze karna hai.

Without Managed Service

Tum EC2 (VMs) lete ho. Fir:

Install Hadoop → Install Spark → Configure Cluster → Manage Nodes → Monitor Resources → Handle Scaling → Perform Upgrades → Handle Failures

Sab tum karoge.

Managed Service (Redshift / BigQuery)

AWS/Google bolta hai: "Data load karo aur queries run karo."

Tum bas use karoge: SELECT * FROM sales;

Then AWS/Google: Cluster manage karenge → Infrastructure maintain karenge → Scaling handle karenge → Query optimization karenge → Monitoring karenge → Failures handle karenge → High availability provide karenge

Tum bas data analysis par focus karoge.

ServiceManaged kya hai?

EC2X (IaaS)

RDS✔ Managed DB

Various Other IaaS (Infrastructure as a Service)
Yahan AWS tumhe raw infrastructure deta hai.
Tumhe bahut control hota hai.

Examples:
EC2 (Virtual Machines)
EBS (Virtual Disks)
VPC
Elastic IP

Service	Managed kya hai?
S3	✔ Managed Storage
Redshift	✔ Managed Analytics
Elastic Beanstalk	✔ Managed App Platform
Lambda	✔ Managed Compute
API Gateway	✔ Managed APIs

React Developer Example

Tum React + Node app bana rahe ho.

Non-managed approach

EC2 -> Install Node-> Install PostgreSQL-> Configure Nginx ->Take Backups ->Monitor Everything. Tumhari responsibility.

Managed approach

Elastic Beanstalk → App

RDS → Database

S3 → File Storage

CloudFront → CDN

Tum bas application logic likhte ho.